From Language to Country to Valence: A Multilingual Audit of Stereotype Bias Across LLMs

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Abstract

Large language models (LLMs) are increasingly considered for education, recruitment, workplace decision support, and other consequential applications, making pre-deployment social-bias detection essential. Yet major benchmarks are English-centred or translated, focus mainly on negative stereotypes, and may miss linguistic and cultural variation. We investigate gender, racial, and religious bias using 12,000 locally written Reddit posts in English, Spanish, French, and German. Only search terms are translated, preserving local context. The dataset is balanced by language and category; English gender is additionally balanced by stereotype valence. Three open-weight and three proprietary API-based models are evaluated with a two-blank design that reduces grammatical-gender cues. Religion has the highest absolute choice imbalance in every model–language condition. Spanish is highest for gender, whereas German is highest for race and religion. The evaluated API models score below the open-weight models throughout. Negative English-gender items exceed positive items; Australian and Mexican contexts exceed their Canadian and Spanish counterparts. These descriptive differences measure choice concentration, not stereotype endorsement or causal country effects. The crossed patterns show how aggregate scores can conceal category- and context-specific behaviour, motivating closer inspection before deployment.

1 Introduction

LLMs increasingly mediate information and decisions in education, recruitment, and workplaces. Their integration into collaborative workflows makes pre-deployment auditing relevant not only to model developers but also to teams selecting and localising decision-support systems (Gallegos et al., 2024; Weidinger et al., 2022; Mitchell et al., 2019; Raji et al., 2020). A system used across languages and communities may exhibit different patterns for different social groups; a global score can obscure the contexts requiring closer scrutiny. Nevertheless, what a bias score establishes depends on its operationalisation (Blodgett et al., 2020; Lum et al., 2025).

Existing evaluations probe associations (Caliskan et al., 2017; May et al., 2019), contrastive sentences and questions (Nangia et al., 2020; Nadeem et al., 2021; Parrish et al., 2022), or generated text (Sheng et al., 2019; Dhamala et al., 2021; Nozza et al., 2021; Smith et al., 2022; Sap et al., 2020). English-dominant resources and translated examples, however, may not preserve local social situations, linguistic forms, or stereotype content (Joshi et al., 2020; Artetxe et al., 2020; Talat et al., 2022; Hershcovich et al., 2022). Cross-cultural and multilingual evaluations demonstrate the importance of language, locale, and variety (Gamboa et al., 2025; Nie et al., 2024; Naous et al., 2024; Fleisig et al., 2024).

We translate retrieval terms rather than complete posts, building on Reddit-based bias evaluation (Barikeri et al., 2021). Locally written contexts preserve native wording and community references, but are deliberately non-parallel: the resulting comparisons describe context-specific model behaviour, not isolated causal language effects. Agreement-aware linked blanks reduce grammatical clues (Zhao et al., 2018; Stanovsky et al., 2019; Zhao et al., 2020). We also separate positive and negative stereotypes: favourable generalisations can still essentialise a group (Fiske et al., 2002; Kay et al., 2013; Siy and Cheryan, 2016).

Our contributions are:

- We construct a 12,000-item multilingual dataset from locally written Reddit posts, balanced across English, Spanish, French, and German and the gender, race, and religion categories. Translating only search terms, followed by multilingual filtering and language-

aware human annotation, preserves locally written context.

- We evaluate three open-weight and three proprietary API-based models with an agreement-aware two-blank format that prevents grammatical gender from revealing the expected answer.
- We add two focused context analyses: positive versus negative gender stereotypes, and within-language country comparisons for English and Spanish.

2 Related Work

CrowS-Pairs and StereoSet evaluate preferences among stereotypical and counter-stereotypical alternatives (Nangia et al., 2020; Nadeem et al., 2021); BBQ examines bias in question answering (Parrish et al., 2022). RedditBias brings naturally occurring conversation into evaluation (Barikeri et al., 2021), while counterfactual probes compare outcomes after changing sensitive attributes (Huang et al., 2020). Such instruments measure distinct behaviours, not interchangeable indicators of downstream harm. Constrained tests need not predict behaviour in longer responses (Lum et al., 2025).

Multilingual audits additionally face translation and cultural-validity problems (Talat et al., 2022; Nie et al., 2024). Region-specific Spanish resources and evaluations distinguish varieties rather than treating a shared language as a single cultural setting (España-Bonet and Barrón-Cedeño, 2024; Grandury et al., 2025). Cross-cultural value probes and cultural-adaptability frameworks likewise motivate context-sensitive evaluation (Arora et al., 2023; Rao et al., 2025). Reddit corpora offer community-grounded material for such work (Baumgartner et al., 2020; Barikeri et al., 2021). Our design complements controlled translation studies by preserving local contexts and reporting crossed category–language profiles. Its non-parallel corpora limit attribution of a difference to language alone.

3 Dataset and Evaluation Design

3.1 Evaluation Questions

RQ1: Does aggregate constrained stereotype-choice imbalance remain stable across languages and models? **RQ2:** How do gender, race, and religion differ across those languages and models? **RQ3:** How does imbalance differ by English-gender stereotype valence and between country-linked contexts within a shared language?

3.2 Local-Language Collection and Allocation

Following RedditBias's distinction between targets and attributes (Barikeri et al., 2021), queries combine identity terms with traits, roles, or behaviours. The project-defined comparisons are women with men, White with Black, and Christians with Muslims; these are tested label pairs, not exhaustive identity taxonomies. Gender attributes include *caring, capable, emotional, domineering*; race attributes include *hardworking, resilient, aggressive, untrustworthy*; religion attributes include *charitable, compassionate, dogmatic, intolerant*. Attributes retrieve candidates, rather than predetermine which group is associated with a trait. Contextual annotation determines stereotype presence and valence. Equal category sizes do not match lexical severity or topic, and alternative-attribute robustness is not tested.

GPT-4o, DeepL, and Google Translate independently produce Spanish, French, and German search-term candidates. Source and candidate terms are encoded with `sentence-transformers/paraphrase-multilingual-mpnet-base-v2`, a multilingual knowledge-distillation checkpoint (Reimers and Gurevych, 2020). Candidates with cross-lingual cosine similarity above .8 are retained; disagreements and morphologically ambiguous terms receive human review. Only the retained terms are translated: retrieved posts remain in their original language and local context.

Each language contributes 1,000 items per category, hence 3,000 per language and 12,000 overall. English and Spanish are each divided equally between two country-linked source strata: Canada/Australia and Mexico/Spain, respectively. Each country contributes 500 items per category. The English-gender cell additionally contains 500 positive and 500 negative items, split 250/250 within each country; other cells have no valence quota. French and German contribute 1,000 items per category without a country split. Country provenance indicates a community context, not an author's nationality. The English frame includes `r/australia`, `r/sydney`, `r/melbourne`, `r/canada`, `r/toronto`, and `r/vancouver`; these communities are not assumed nationally representative.

Language	R	D	L	C	U	F
English	14,808	12,868	11,353	6,733	3,615	3,000
Spanish	14,193	12,346	11,028	7,527	3,574	3,000
French	12,642	11,931	10,554	5,348	4,521	3,000
German	13,976	11,187	10,763	6,412	5,548	3,000
Total	55,619	48,332	43,698	26,020	17,258	12,000

Table 1: Multilingual collection flow. R/D/L/C/U/F denote retrieval, deduplication, language pass, context pass, unanimous annotation, and final allocation.

3.3 Filtering and Human Annotation

We remove deleted content, very short posts, duplicates, quotations, counter-speech, and ambiguous cases; verify language and community provenance; and require a meaningful target–attribute relation. Table 1 reports each filtering stage. Annotators are recruited through Amazon Mechanical Turk (MTurk) and must meet the task's qualification criteria. Three native-language annotators judge each remaining candidate. Country-conditioned items use annotators resident in, and familiar with, the corresponding country; country eligibility is checked during qualification. Before annotation, workers receive the participant information and consent statement reproduced in Appendix A.

One multiple-choice selection combines stereotype presence, category, and valence: no stereotype, or positive/negative gender, race, or religion. Retention requires unanimous agreement on the same non-zero label. Positive labels are available in every language, but only English gender is quota-balanced for the valence study. Retained counterfactuals are checked for grammaticality, semantic equivalence apart from identity, and predefined stereotype direction.

3.4 Agreement-Aware Prompting and Models

For race and religion, counterfactuals change the identity while retaining the surrounding sentence. Gender agreement requires linked changes: "Ella es ___" can reveal *ingeniera* through grammar rather than association. We instead blank both subject and occupation, for example "___ trabaja como ___ en una empresa de tecnología", with alternatives *Él/ingeniero* and *Ella/ingeniera* (he/engineer and she/engineer). Both alternatives are grammatical. The pair is swapped as a unit and A/B option order is counterbalanced.

The six systems are LLaMA 3 8B, Mistral 7B, and Gemma 7B (open-weight), and GPT-4o, Claude 3 Opus, and Gemini 1.5 Pro (proprietary APIs). Decoding uses temperature .7, top-p .9, short outputs, randomised option order, and ten valid categorical responses per prompt. The protocol records raw responses, parsed choices, refusals, invalid answers, and retry indices separately. The score conditions on valid answers; refusal rates are not included in it. Model access is not a controlled factor: scale, training data, alignment, and serving procedures also differ.

4 Measurement and Descriptive Scope

For item i, let X_i count selections of the context-annotated stereotypical alternative among k valid responses. We report

$$B_{i,k} = \frac{2|X_i - k/2|}{k}, \qquad \overline{B}_k = \frac{1}{N}\sum_{i=1}^{N} B_{i,k}. \tag{1}$$

At $k = 10$, item scores are $0, .2, .4, .6, .8, 1$. Zero is an observed five–five split; one is unanimity. Ten stereotypical and ten counter-stereotypical choices both score one. Thus B measures *choice concentration*, not endorsement direction or social harm.

Even independent balanced responses yield non-zero absolute scores. For $X \sim \mathrm{Binomial}(k, .5)$,

$$\mathbb{E}_0[B_k] = \frac{2}{k2^k}\sum_{x=0}^{k}\left|x - \frac{k}{2}\right|\binom{k}{x}. \tag{2}$$

For ten repetitions, this equals $63/256 = .24609375$, and $P_0(B_{10} > 0) = .75390625$. The reference depends on repetition count and is diagnostic, not a calibrated test of dependent model responses. Being above or below it establishes neither preference direction nor statistical significance.

All rankings and contrasts below are descriptive. Sample SDs indicate dispersion across the specified models or model-paired differences, not item-level

Model	English	Spanish	French	German
LLaMA 3	.308/.31/.33	.39/.39/.41	.32/.36/.40	.36/.41/.43
Mistral 7B	.285/.28/.29	.36/.36/.38	.29/.33/.36	.33/.37/.40
Gemma 7B	.345/.34/.35	.42/.42/.44	.35/.39/.42	.39/.44/.46
GPT-4o	.1795/.18/.19	.23/.22/.25	.18/.21/.24	.21/.24/.27
Claude 3	.172/.17/.18	.22/.21/.24	.17/.20/.23	.20/.23/.26
Gemini 1.5	.1915/.19/.20	.24/.23/.26	.19/.22/.25	.22/.25/.28
Mean	.247/.245/.257	.310/.305/.330	.250/.285/.317	.285/.323/.350
SD	.075/.074/.076	.090/.095/.090	.079/.085/.086	.085/.094/.090

Table 2: Core absolute-imbalance results. Each cell lists gender/race/religion. The mean and sample SD rows summarise six models per language–category cell; SD measures between-model dispersion, not item-level uncertainty. English gender is pooled from the equal valence splits in Table 3.

sampling uncertainty. The six systems are a fixed evaluation set; common rankings across them do not establish generalisation beyond the sampled items or model set. Summaries are calculated from tabulated condition values before display rounding; last-digit differences should not be overinterpreted.

5 Results and Interpretation

5.1 Category, Language, and Model Profiles

The core crosses six models, four languages, and three categories. Table 2 retains all 72 condition scores. Category-averaged language means are .250 for English, .284 for French, .315 for Spanish, and .319 for German (RQ1). English is lowest for every model; German is highest or tied highest at the tabulated precision. Across the 24 model–language cells, religion averages .313 ($SD = .087$), race .290 ($SD = .087$), and gender .273 ($SD = .081$). Religion is highest in every cell; every model retains this category order after averaging languages (RQ2). These are subcorpus-dependent concentration rankings, not a hierarchy of social harm.

5.1.1 Within-language category contrasts

English follows religion $>$ gender $>$ race (.257/.247/.245), whereas Spanish has religion $>$ gender $>$ race (.330/.310/.305). Its gender–race separation is only .005. French and German return to religion $>$ race $>$ gender, with religion–gender gaps of .067 and .065, compared with .010 for English. English has the narrowest category range (.012), compared with .025 for Spanish. A language-wide mean therefore mixes distinct category profiles rather than describing equally elevated scores for every group.

The French profile also persists within access families. Open-weight gender/race/religion means are .320/.360/.393; API means are .180/.210/.240. French gender averages .320 ($SD = .030$) among the three open-weight models and .180 ($SD = .010$) among the APIs. The same category ordering survives a substantial difference in model baselines. Between-family separation partly explains the larger six-model SDs and should not be mistaken for uncertainty over items.

5.1.2 Within-category language contrasts

For gender, the six-model means order English $<$ French $<$ German $<$ Spanish. English and French are nearly tied (.247 versus .250): Claude and Gemini have slightly higher English than French gender scores at the tabulated precision. For race and religion, every model orders English $<$ French $<$ Spanish $<$ German. Spanish exceeds German by .025 for gender, while German exceeds Spanish by .018 for race and .020 for religion. Their .004 aggregate gap thus hides a crossover: neither language is uniformly higher. Reading categories separately changes which local contexts appear most concentrated, despite the common aggregate language ordering.

Every evaluated open-weight score exceeds every API score in its matched cell, but family gaps range from .130 for English race to .170 for Spanish race. The separation is therefore neither constant across languages nor shared equally by categories. These are observed access-family patterns, not effects of being "free" or "paid".

5.1.3 Interpreting the finite-sample reference

From the tabulated scores, all 12 open-weight model–language means exceed .2461; 11 of the 12 API means are below it. The exception is Gemini 1.5 Pro in German (.250). This comparison locates the point estimates, not which cells differ statistically from balanced choice. Below-reference scores do not imply counter-stereotypical prefer-

Model	Core	Positive	Negative	Δ_{val}
LLaMA 3 8B	.368	.247	.369	+.122
Mistral 7B	.336	.233	.337	+.104
Gemma 7B	.397	.274	.416	+.142
Open-weight mean	.367	.251	.374	+.123
GPT-4o	.217	.147	.212	+.065
Claude 3 Opus	.207	.143	.201	+.058
Gemini 1.5 Pro	.227	.158	.225	+.067
API mean	.217	.149	.213	+.063

Table 3: Model-level absolute imbalance and English-gender valence results. Core averages the tabulated language–category scores; $\Delta_{\text{val}} = B_{\text{negative}} - B_{\text{positive}}$.

ence and warrant checking repetition dependence; above-reference scores likewise do not establish stereotype endorsement.

5.2 Study 1: English-Gender Stereotype Valence

For RQ3's valence comparison, positive items average .200 ($SD = .058$) and negative items .293 ($SD = .092$). Table 3 gives the six model results. Model-paired negative-minus-positive differences average .093 ($SD = .035$), range from .058 to .142, and are positive for every model. The open-weight mean contrast (.123) exceeds the API contrast (.063).

Higher negative-item scores describe stronger concentration, not greater endorsement of negative stereotypes. Positive and negative examples are different posts, balanced in count within each country, rather than an intervention reversing the valence of otherwise identical sentences. The common sign across models does not establish item-level precision. The result is restricted to English gender; the potential harm of favourable generalisations is a distinct concern (Fiske et al., 2002; Siy and Cheryan, 2016).

5.3 Study 2: Country Context Within a Shared Language

Country comparisons retain the language, models, category allocation, decoding procedure, and score definition while changing the country-linked Reddit source. Table 4 reports the category-specific scores. Categories are first averaged within each model for locale-level summaries. Canada averages .220 ($SD = .069$), versus .279 ($SD = .081$) for Australia; their model-paired difference is .059 ($SD = .013$). Spain averages .301 ($SD = .088$), versus .329 ($SD = .095$) for Mexico; the paired difference is .028 ($SD = .007$).

The Australia–Canada gender/race/religion contrasts are .094/.036/.048, with paired SDs .018/.009/.011. Mexico–Spain contrasts are .018/.028/.038, with SDs .005/.007/.009. Every model's contrast is positive in every category, but the largest gap is gender for Australia–Canada and religion for Mexico–Spain. Canada follows religion $>$ race $>$ gender, whereas Australia follows gender $>$ religion $>$ race (.294/.281/.263). Spain places gender above race, whereas Mexico's gender and race means are both .319. Thus a shared direction at country level can still combine different category profiles within the same language.

Open-weight mean country gaps exceed API gaps: .070 versus .049 for Australia–Canada and .034 versus .022 for Mexico–Spain. Larger SDs for country scores than for paired differences reflect differing model baselines, not precise estimation over Reddit items. Pairing is between model summaries; posts from different countries are not matched sentence pairs. Topic, community, and wording vary alongside country context, so these findings neither identify causal country effects nor rank national populations by prejudice.

6 Discussion and Deployment Implications

The Spanish–German crossover is the central example of aggregation hiding context: a small language-wide difference combines opposing category orders. The French access-family comparison and the country-specific category profiles similarly show why a single model or language summary is incomplete. For teams designing multilingual decision-support workflows, these descriptive profiles identify context–category cells that merit closer inspection, rather than establish a universal model ranking.

A practical implication is to document the intended languages, communities, and social categories when comparing candidate models, and to examine representative responses and refusals alongside concentration scores. This links local evaluation to model reporting and internal audit processes (Mitchell et al., 2019; Raji et al., 2020). It is a deployment implication, not a tested intervention: this study does not evaluate a deployed collaborative system or show improved human decision outcomes. Lower absolute scores do not certify fairness; high scores can reflect consistently

Locale	Category	LLaMA	Mistral	Gemma	GPT-4o	Claude	Gemini	Mean ± SD
English: Canada–Australia								
Canada	Gender	.255	.237	.283	.141	.132	.150	.200 ± .066
	Race	.288	.260	.316	.166	.157	.175	.227 ± .069
	Religion	.301	.263	.319	.171	.162	.180	.233 ± .070
Australia	Gender	.361	.333	.407	.218	.212	.233	.294 ± .084
	Race	.332	.300	.364	.194	.183	.205	.263 ± .079
	Religion	.359	.317	.381	.209	.198	.220	.281 ± .081
Spanish: Spain–Mexico								
Spain	Gender	.379	.350	.408	.223	.214	.232	.301 ± .088
	Race	.373	.344	.402	.209	.200	.218	.291 ± .092
	Religion	.387	.358	.416	.235	.226	.244	.311 ± .085
Mexico	Gender	.401	.370	.432	.237	.226	.248	.319 ± .092
	Race	.407	.376	.438	.231	.220	.242	.319 ± .099
	Religion	.433	.402	.464	.265	.254	.276	.349 ± .094

Table 4: Country-context absolute-imbalance results. Final column: mean and sample SD across six models. Each country-pair midpoint recovers the pooled category cell in Table 2. Australian gender is obtained by equal-country balance from pooled English gender and the Canadian row.

stereotypical or counter-stereotypical choices.

7 Conclusion

We presented a multilingual, context-aware audit across six systems, four languages, and gender, race, and religion. Native posts retain local context and linked blanks reduce grammatical cues. Religion has the highest pooled score in every model–language cell, but Spanish leads for gender whereas German leads for race and religion. API scores are lower than open-weight scores; negative English-gender items and Australian/Mexican contexts have higher observed concentration than their counterparts. Australia's gender-led profile contrasts with Canada's religion-led profile. These descriptive patterns support category- and context-specific inspection, not conclusions about endorsement direction or causal country effects. In consequential collaborative workflows, examining actual responses before acting on aggregate scores can help avoid unsupported fairness assurances affecting people's opportunities and treatment.

Limitations

Reddit is not representative of all speakers or deployment settings. Community provenance does not establish nationality, and topic, period, moderation, and demographics can differ. Native-context comparisons cannot isolate language or country effects. Binary gender labels and White–Black and Christian–Muslim pairs omit substantial identity diversity. Category ordering depends on the selected attributes and retained subcorpora; no alternative-lexicon or topic-matched robustness experiment is reported.

Absolute scores discard preference direction, and item-level confidence intervals are not estimated. Rankings are descriptive, not tests of population differences. Repetition dependence and conditioning on valid responses further limit the independent binomial reference. Access-family comparisons confound model scale, training, alignment, and serving.

Ethical Considerations

Before annotation, MTurk workers received information about the study, including a warning that excerpts could contain hostile or favourable stereotypes and potentially distressing material. The statement described voluntary participation, the option to decline answers or discontinue without penalty, compensation through MTurk for completed tasks, and academic-research use of the annotations. Consent was indicated by proceeding after this information was presented. Appendix A reproduces the statement used before annotation. This study received institutional ethics approval; identifying approval details are withheld for anonymous review.

Annotator identifiers were removed and annotation results reported in aggregate. Other identifiers were removed and verbatim text minimised. Public visibility is not blanket permission to republish Reddit content, and the paper's licence does not confer rights over third-party posts (Reddit, 2026). No unrestricted raw-text release is provided. Any

release of prompts, annotations, response counts, or code requires explicit access terms, privacy safeguards, and applicable retention and deletion procedures.

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A MTurk Participant Information and Consent

The following statement was presented to MTurk workers before they annotated the data reported in this paper. Its wording is reproduced here, with formatting adapted to the paper layout.

Participant Information and Ethical Statement

You are invited to participate in an academic research study examining how large language models associate social groups with different characteristics across languages and cultural contexts.

As part of this task, you will be asked to review short excerpts of publicly available Reddit posts and classify whether they contain stereotypes relating to gender, race, or religion. Some materials may express negative or hostile stereotypes, while others may contain seemingly favourable or positive stereotypes. These materials are presented solely for research and evaluation purposes and do not represent the views or beliefs of the researchers.

Potential risks and discomfort: Some excerpts may contain sensitive, offensive, or potentially distressing content involving social identities. Please be aware of this before proceeding. You may decline to answer or discontinue participation if you feel uncomfortable.

Voluntary participation: Your participation is entirely voluntary. You are free to stop participating at any time, without penalty.

Compensation: You will receive compensation through Amazon Mechanical Turk for completing the assigned annotation task.

Privacy and confidentiality: Your personal identity will not be disclosed in any research publication. Annotator identifiers are removed from the research dataset, and results are reported in aggregate. The research focuses on annotation judgements rather than the personal opinions, beliefs, or identities of individual participants.

Data use: The annotations collected will be used exclusively for academic research on multilingual stereotype patterns and the evaluation of large language models. The research findings may be published in academic papers and related research outputs. Individual annotators will not be identified in these publications.

By proceeding with this task, you acknowledge that you have read and understood this information, are aware of the potentially sensitive nature of the materials, and voluntarily agree to participate in the annotation study.